FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2021

TEST DATE: 16-05-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. Net force on system is zero so acceleration of centre of mass will zero.

2. A

3. B

Sol. $T_1 = 2\pi\sqrt{\frac{\ell}{g}}$ .....(1)

$T_2 = 2\pi\sqrt{\frac{\ell}{g'}}$ .....(2)

$g' = \frac{gR^2}{(R+2R)^2} = \frac{g}{9}$

$\Rightarrow T_2 = 2\pi\sqrt{\frac{9\ell}{g}} = 3T_1$

$\Rightarrow \frac{T_1}{T_2} = \frac{1}{3}$

4. B

Sol. $\frac{-GMm}{R} + \frac{1}{2}m\frac{9}{16}\frac{2GM}{R} = \frac{-GMm}{R+h}$

$$\Rightarrow \frac{-7}{16}\frac{GMm}{R} = \frac{-6Mm}{R+h}$$

$$\Rightarrow h = \frac{9R}{7}$$

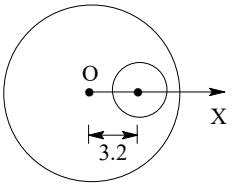
5. A

Sol. $a_{max} = w^2 A$

$$a_{max} = 100\left(\frac{10}{100}\right) = 10\, m/s^2$$

6. A

Sol. Mass of removed disc $= \frac{m}{\pi 6^2}\pi 2^2 = \frac{m}{9}$



$$X_{cm} = \frac{m\times O + \left(\frac{-m}{9}\right)\times 3.2}{m+\left(\frac{-m}{9}\right)} = \frac{-3.2/9}{8/9} = -0.4\,\text{cm}$$

7. D

Sol. $P = \frac{mgh}{t}$

$$4\times 10^3 = \frac{m(10)(10)}{60}$$

$$\Rightarrow m = 2400\, Kg$$

$$\Rightarrow V\times 1000 = 2400$$

$$\Rightarrow V = 2.4\, m^3$$

$$\Rightarrow V = 2400\, litre$$

8. A

Sol. $W = m\times area\, under\, the\, curve$

$$W = 5\left(\frac{1}{2}\times 8\times 20\right)\times 10^{-4}$$

$$W = 4\times 10^{-2} J$$

9. C

Sol. Length of air column on both side is 45 cm when one side at 0°C and the other is at 273°C. The pressure must be same on both sides. Hence

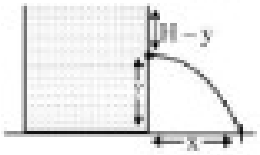
$$\frac{l_1}{T_1} = \frac{l_2}{T_2} \Rightarrow \frac{l_1}{273} = \frac{l_2}{(273+273)} \Rightarrow l_1 = \frac{l_2}{2}$$

Applying gas equation to the side at 0°C, we get

$$\frac{P_1 l_2}{T_1} = \frac{Pl}{T} \Rightarrow \frac{P_1 \times 30}{273} = \frac{76 \times 45}{273 \times 81} \Rightarrow P_1 = 102.4$$

10. C

Sol. Time taken: $y = \frac{1}{2} g t^2 \Rightarrow t = \sqrt{\frac{2y}{g}}$



$$x = ut = \sqrt{2g(H-y)}\left(\sqrt{\frac{2y}{g}}\right)$$

$$= 2\sqrt{y(H-y)} = 12cm$$

11. B

Sol. $$\frac{dx}{dt} = \sqrt{\frac{\mu x g}{\mu}}$$

$$\int_0^{\ell} X^{1/2} dx = \sqrt{g} \int_0^t dt$$

$$t = 2\sqrt{\frac{\ell}{g}} = 2\sqrt{\frac{2.45}{9.8}} \Rightarrow t = 1\sec$$

12. B

Sol. $$L = 10 \log \frac{I}{I_0}$$

$$60 = 10 \log \frac{I}{10^{-12}}$$

$$I = 10^{-6}\, W/m^2$$

$$P = 10^{-6} \times 4\pi (500)^2$$

$$10^3 \times \frac{30}{100} = 4\pi \times 25 \times 10^{-2} t$$

$$t = \frac{300}{\pi} = 95.5\sec$$

13. A

Sol. Time taken by stone to reach floor t = h/u. To find the speed of Iron man we can use momentum conservation

$$Mv_1 = mu$$

$$v_1 = \frac{mu}{M}$$

Distance of Iron man moved above platform is $y=\left(\frac{mu}{M}\right)\left(\frac{h}{u}\right)=\frac{mh}{M}$

Total height $=h\left(1+\frac{m}{M}\right)$

14. A

15. C

16. B

Sol. $W=\int 2y\ dx+\int x^2dy \qquad here\ y=x$

$W=\int_0^5 2xdx+\int x^2dx$

$W=W=25+\frac{125}{3}=\frac{200}{3}=66.7J$

17. A

Sol. Distance $AD=\frac{4}{5}\times 100=80\,cm$

18. A

Sol. $\frac{N_1}{N_2}=\frac{N_0e^{-6\lambda t}}{N_0e^{-4\lambda t}}$

$\frac{1}{e}=\frac{1}{e^{2\lambda t}}\Rightarrow 2\lambda t=1$

$t=\frac{1}{2\lambda}\sec$

19. A

Sol. From ohm's law

Electric field α current density

20. A

SECTION – B

21. 00000.60

Sol. $v=a\omega$

$0.6=a\times 10$

$a=0.06m$

22. 00005.00

Sol. $K\frac{20}{100}=\frac{K10}{100}+mg$

$K=10mg$

$$T = 2\pi\sqrt{\frac{m}{K_1 + K_2}}$$

$$T = 2\pi\sqrt{\frac{m}{2K}} = 2\pi\sqrt{\frac{3}{20mg}}$$

$$T = \frac{\pi}{5}\sqrt{\frac{1}{2}} \Rightarrow N = 5$$

23. 00000.64

24. 00012.00

Sol. $$\frac{\Delta x}{x} = \pm\left(\frac{2\Delta a}{a} + \frac{3\Delta b}{b} + \frac{\Delta C}{C} + \frac{\Delta d}{2d}\right)$$

$$= \pm\left(\left(2\times 2\right) + \left(3\times 1\right) + 3 + \left(\frac{1}{2}\times 4\right)\right)$$

$$= \pm 12\%$$

25. 00141.40

Sol. Conservation of volume $\left(A - a\right)y = ax$

So, $a\left(y + x\right)\rho g = mb$

$$b = \frac{a\rho g}{m}\left(\frac{ax}{A-a} + x\right)$$

$$b = \frac{a\rho g\left(ax + Ax - ax\right)}{m\left(A-a\right)}$$

$$b = \frac{aA\rho g\ x}{m\left(A-a\right)} \Rightarrow \omega = \sqrt{\frac{Aa\rho g}{m\left(A-a\right)}} = 141.40$$

26. 00001.75

Sol. $V_1 = Cq/a \qquad V_2 = C'6q/a\ \left(Assume\ a\ cube\ of\ side\ 2a\right)$

$$V_1^1 = \frac{3C4q}{2a}\times 2 \qquad V_2^1 = \frac{C^1 24q}{2a}$$

$$V^1 = V_1^1 + V_2^1 = \frac{3}{2}V_1 + \frac{V_2}{4}$$

$$a = \frac{3}{2}\ \&\ b = \frac{1}{4}$$

So, $a + b = 1.75$

27. 00065.67

Sol. Time period of a satellite in circular orbit is given by

$$T = 2\pi\sqrt{\frac{r^3}{GM}}; dT = 1.62\,hr \ = \frac{2\pi}{\sqrt{GM}}\times\frac{3}{2}r^{1/2}dr$$

$$dT = \frac{3}{2} \times \frac{70}{7000} \times T$$

$$= 0.015T$$

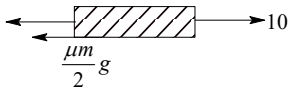
Let the first satellite catch up with the second after n revolutions, then

$$nT = (n+1)(T - dT) \Rightarrow \frac{n}{n+1} = 0.985$$

$$n = 65.67$$

28. 00002.50

Sol. $T = 10 - \frac{3}{4} \times 10$



$$T = \frac{5}{2} N = 2.5$$

29. 00000.50

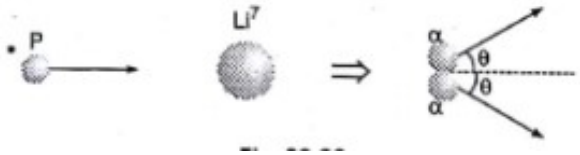
Sol. $T = 2\pi\sqrt{\frac{I}{mgl}}$

$$\Rightarrow 2 = 2\pi\sqrt{\frac{2mr^2}{mgr}} = 2\pi\sqrt{\frac{2r}{g}}$$

$$\Rightarrow r = \frac{1}{2} = 0.5m$$

30. 00017.28

Sol. $Q = (2 \times 4 \times 7.06 - 7 \times 5.6)\,\text{MeV} = 17.28\,\text{MeV}$



Applying conservation of energy for collision, $K_p + Q = 2K_\alpha$

$$\sqrt{2m_p K_p} = 2\sqrt{2m_\alpha K_\alpha}\cos\theta$$

$$K_p = 16K_\alpha \cos^2\theta = (16K_\alpha)\left(\frac{1}{4}\right)^2$$

Solving eqs. (1) and (3) with $Q = 17.28\,\text{MeV}$

We get $K_p = 17.28\,\text{MeV}$

Chemistry

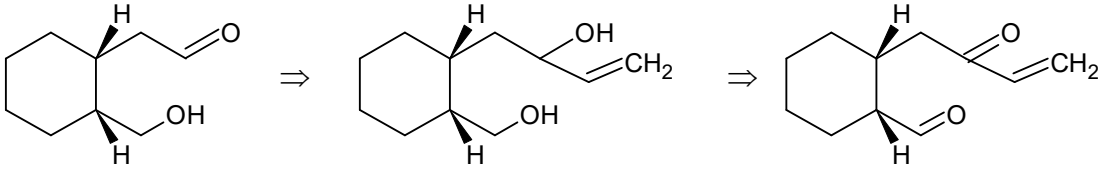
PART – B

SECTION – A

31. D
Sol. Theoretical

32. D
Sol. $S=C=C=C=S$

33. B
Sol.



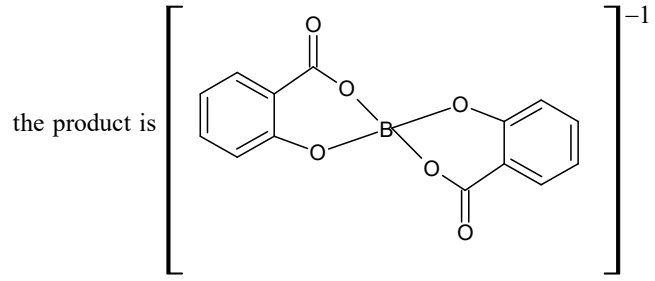
34. B
Sol. (D) is $_{71}Lu$

35. C
Sol. Standard boiling point = 99.6°C
Normal boiling point = 100°C

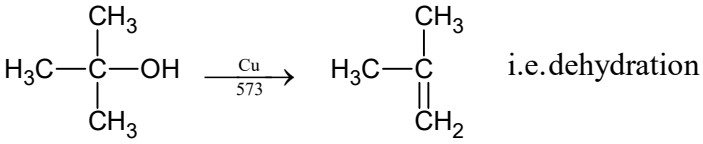
36. A
Sol. $C(s)+2Cl_2(g)\longrightarrow CCl_4(\ell) \qquad \Delta H=-135.5\,KJ/mole$
$CCl_4(\ell)\longrightarrow CCl_4(g) \qquad \Delta H=30.5\,KJ/mole$
$C(s)+2Cl_2(g)\longrightarrow CCl_4(g) \qquad \Delta H=-105\,KJ/mole$
$715+2\times 242-4\times x=-105$
$x=326\,KJ/mole \qquad i.e.\,x=B.E.of\ C-Cl\ in\ CCl_4(g)$

37. D
Sol. The total energy of the system takes into account the kinetic energies of all the sub-atomic particles, attractive potential between the electrons and nuclei also **include** repulsive potential among the electrons and nuclei individually.

38. B
Sol.



39. D

Sol.  i.e. dehydration

40. B

Sol. Polymer is prepared by condensation reaction and cross link polymer.

41. D

Sol. Antibiotics have either cidal (killing) effect or static (inhibitory) effect on microbes

42. A

Sol. Vitamin B_1=Thiamine

43. D

Sol. VSEPR theory, for B part $\alpha \neq 120°$ due to repulsion of π electron

44. A

Sol. $Ni(OH)_2(s) \rightleftharpoons Ni^{2+} + 2OH^-$

$s \qquad 2s + 0.1$

$s(2s+0.1)^2 = 2\times10^{-15} \qquad 2s + 0.1 \approx 0.1$

$s = 2\times10^{-13}$

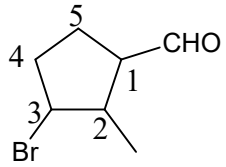
45. B

Sol. The process is non- spontaneous at given condition so $\Delta G > 0$

46. D

Sol. Malachite is $CuCO_3.Cu(OH)_2$

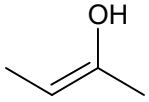
47. C

Sol. 

48. B

Sol. One M^{3+} ion can replace three Na^{+1} ion so the cationic vacancies produced are equal to twice the number of M^{3+} ions.

49. C

Sol.  this enol form is more stable

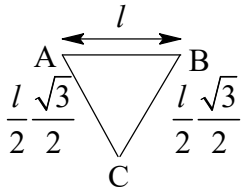
50. B

Sol. ZnS is white coloured ppt which is soluble in acidic solution below pH=6

SECTION – B

51. 00005.60

Sol.



where A and B are tetrahedral voids and C is octahedral void

52. 00004.00

Sol. i, iii, iv, v

53. 00003.00

Sol. 20 mL Au sol, 1 ml 10% NaCl requires = $6\times10^{-5}gm$ protective colloid

10 mL Au sol, 1 ml 10% NaCl requires = $3\times10^{-5}gm$ protective colloid

$=3\times10^{-2}mg$

54. 00009.00

Sol. $CFSE = 2PE-\left(2PE-5\times0.4\times\Delta_0+2\times0.6\Delta_0\right)$

$7200 = 0.8\Delta_0$

$\Delta_0 = 9000$

55. 00008.00

Sol. Compound is Urotropin $\left(CH_2\right)_6 N_4$ x+y=4+4=8

56. 00005.00

Sol. $\Delta U=q+w$

$\Delta U = 0+w$

$$2\times C_V\times\left(T_2-350\right) = -2\left[\frac{\left(2\times R\times T_2\right)}{\frac{8}{3}}-700R\right]$$

$T_2=600$

$$W=-2\left[\frac{\left(2\times R\times 600\right)}{\frac{8}{3}}-700R\right]=500R$$

57. 00001.00

Sol. Here on dilution pH is very slightly effected.

58. 00004.00

Sol. $O_2+2H_2O+4e^-\longrightarrow 4OH^-$

59. 00006.00

Sol. $P_{mix} = 6.0 + 1.5 X_B$

$P_A^* = 6 + 1.5 \times 0$

$= 6$

60. 00003.00

Sol. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Initial mole 1 3 0

At eq. 1 – x 3 – 3x 2x

∵ Out of 4 mole of mixture, 2 mole have reacted and hence 2 mole are left.

∴ 1 – x + 3 – 3x = 2

Or 4x = 2 or x = 0.5

∴ Total mole at equilibrium = 1 – x + 3 – 3x + 2x = 4 – 2x = 4 – 2 × 0.5 = 3

No. of mole of NH_3 at equilibrium = 2x = 2 × 0.5 = 1

$\therefore p_{NH_3} = \frac{1}{3} \times P = \frac{P}{3}$

∴ y = 3.

Mathematics

PART – C

SECTION – A

61. D

Sol. $f(x)=x^4+2x^3-x^2+1$ is polynomial of even degree hence its range can't be *R*.

Hence not surjective.

$f(x)=x^3+x+1,\ f'(x)=3x^2+1 \Rightarrow$ monotonic hence bijective.

$f(x)=\sqrt{1+x^2}$, neither surjective not injective

$f(x)=x^3+2x^2-x+1,\ f'(x)=3x^2+4x-1 \Rightarrow$ $D = 16 + 12 > 0$.

Hence $f(x)$ is non-monotonic cubic polynomial hence surjective but not injective.

62. B

Sol. For $0<x<\frac{\pi}{2}$ or $\frac{\pi}{2}<x<\pi$, $0 < \sin x < 1$.

$\therefore$ for $n > 1$, $\sin x > \sin^n x$

$\therefore \quad f(x)=\left[\frac{3(\sin x-\sin^n x)}{\sin x-\sin^n x}\right]=3$, $x\neq\frac{\pi}{2}$ and $f(x) = 3$, $x=\frac{\pi}{2}$

Thus in $(0, \pi)$, $f(x) = 3$.

Hence $f(x)$ is continuous and differentiable at $x=\frac{\pi}{2}$.

63. C

Sol. If $x_0 = K \in I$, then $\lim\limits_{x\to K+} f(x)=\lim\limits_{x\to K} K+\sqrt{x-K}=K$

and $\lim\limits_{x\to K-} f(x)=\lim\limits_{x\to K-}(K-1)+\sqrt{x-(K-1)}$

$=\lim\limits_{x\to K}(K-1)+\sqrt{x-K+1}=K-1+1=K$

Hence *f* is continuous at every $x_0 = K \in I$. If $x_0 \in R \sim I$, then [*x*] is continuous at x_0 which is turn gives that *f* is continuous at every $x_0 \in R \sim I$

64. A

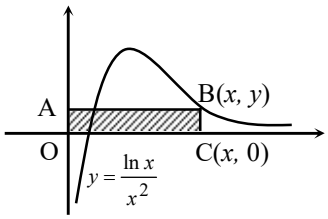
Sol. Area = $A = xy$

$A=\frac{\ln x}{x} \Rightarrow \frac{dA}{dx}=\frac{1-\ln x}{x^2}$

$\frac{dA}{dx}=0 \Rightarrow x = e$

$\frac{dA}{dx}<0,\ \forall x>e$ and $\frac{dA}{dx}>0,\ \forall x<e$.

Hence $x = e$ is point of maxima. $A_{\max}=e.\frac{1}{e^2}=\frac{1}{e}$



65. B

Sol. $f(x)=\int\frac{x^2(\sqrt{1+x^2}-1)}{(1+x^2)(1+x^2-1)}dx=\int\frac{\sqrt{1+x^2}-1}{1+x^2}dx=\int\frac{dx}{\sqrt{1+x^2}}-\int\frac{dx}{1+x^2}$

$=\log(x+\sqrt{1+x^2})-\tan^{-1}x+k$

$\therefore\ f(0)=\log 1-\tan^{-1}0+k=k=0$

$\therefore\ f(x)=\log(x+\sqrt{1+x^2})-\tan^{-1}x$

$\therefore f(1)=\log(1+\sqrt{2})-\frac{\pi}{4}$

66. A

Sol. $p\to q$ must be true & q→r must be false

$\therefore$ q must be true, r must be false but p can be true or false.

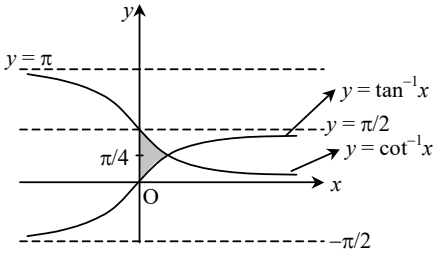
67. D

Sol. Required area,

$\Delta=\int_0^{\pi/4}\tan y\,dy+\int_{\pi/4}^{\pi/2}\cot y\,dy$

$=\ln\sec y\big|_0^{\pi/4}+\ln\sin y\big|_{5\pi/4}^{\pi/2}$

$=\ln\sqrt{2}-\ln\frac{1}{\sqrt{2}}=\ln 2\,.$



68. C

Sol. Put $\tan y=z$

$\therefore\ \sec^2 y\frac{dy}{dx}=\frac{dz}{dx}$

$\therefore\ \frac{dz}{dx}+xz=x^3$

which is a linear differential equation, on solving we get

$\tan y=x^2-2+ce^{-x^2/2}$

69. B

Sol. Numerator = $\cos 6x+\cos 4x+5(\cos 4x+\cos 2x)+10(\cos 2x+1)$

= $2\cos x\{\cos 5x+5\cos 3x+10\cos x\}$

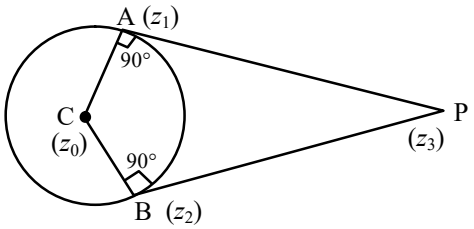
$\therefore y=\frac{\text{Nr}}{\text{Dr}}=2\cos x,\ \frac{dy}{dx}=-2\sin x$

70. C

Sol. Let z_1, z_2 are represented by *A, B* whereas

z_0 is represented by *C*.

Let *P* represent z_3



$$\frac{z_3 - z_1}{z_0 - z_1} = \frac{PA}{AC} e^{\frac{i\pi}{2}} ; \frac{z_0 - z_2}{z_3 - z_2} = \frac{BC}{PB} e^{\frac{i\pi}{2}}$$

$$\frac{z_3 - z_1}{z_0 - z_1} . \frac{z_0 - z_2}{z_3 - z_2}$$

$$= \frac{PA}{\text{radius}} . \frac{\text{radius}}{PB} e^{i\pi} = -1 \ (\because PA = PB)$$

71. B

Sol. Let A_1 be the property that HIN appears

Let A_2 be the property that DUS appears

Let A_3 be the property that TAN appears

$n(A_1) = 7! = 5040$

$n(A_2) = \frac{7!}{2!} = 2520$

$n(A_3) = 7! = 5040$

$n(A_1 \cap A_2) = n(A_2 \cap A_3) = n(A_3 \cap A_1) = 5! = 120$

$n(A_1 \cap A_2 \cap A_3) = 3! = 6$

$n(A_1 \cup A_2 \cup A_3) = \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + n(A_1 \cap A_2 \cap A_3)$ = 12246

Total number of ways to arrange the letters of the word HINDUSTAN $= \frac{9!}{2!}$ = 181440

$\therefore$ Number of ways in which neither the pattern HIN or DUS and TAN appears
= 181440 – 12246 = 169194

72. A

Sol. Total number of matrices = 3^9
Number of symmetric matrices = 3^6
Number of skew symmetric matrices = 3^3
The zero matrix is both symmetric and skew-symmetric the required probability is

$$= \frac{3^9 - 3^6 - 3^3 + 1}{3^9}$$

73. A

Sol. LHS achieves its maximum value at $x = 1$ (in the domain of the equation i.e., $x > 0$)
Maximum value of LHS =1 at $x = 1$
RHS achieves it minimum value at $x = 1$ (in the domain of equation i.e., $x > 0$)
Minimum value of RHS = 1 at $x = 1$
Hence only possible value of x satisfying the given equation is 1.

74. D

Sol. $(1 - x + x^2)^n = a_0 - a_1 x + a_2 x^2 - .... + a_{2n} x^{2n}$

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + ... + {}^nC_n x^n$

On multiplying and collecting coefficient of x^{3r}, we get $\sum_{k=0}^{3r} (-1)^k a_k \ {}^nC_{3r-k}$ = ${}^nC_r$

75. A

Sol. We can write Δ as

$$\Delta = \begin{vmatrix} 0 & \sin x - \cos x & \log\tan x \\ -(\sin x - \cos x) & 0 & -\log\tan x \\ -\log\tan x & \log\tan x & 0 \end{vmatrix}$$

Which is determinant of a skew symmetric matrix of odd order, hence value will be zero

76. B

Sol. $A\,(\text{Adj } A) = |A|\,I_3$

$K = |A| = 2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x)$

$= 2(1 - 3\sin^2 x.\cos^2 x) - 3\,(1 - 2\sin^2 x\cos^2 x) = -1$

77. C

Sol. By the given condition $(1-ab)m^2 - (a^2+b^2)m - (1+ab) = 0$

$\Rightarrow m(a^2+b^2) + (m^2+1)ab = m^2 - 1$ ...(i)

Now H_1 = First H.M. between *a* and *b* = $\dfrac{(m+1)ab}{a+mb}$ and $H_m = \dfrac{(m+1)ab}{b+ma}$

$\therefore\ H_m - H_1 = (m+1)ab\left[\dfrac{1}{b+ma} - \dfrac{1}{a+mb}\right] = (m+1)ab\dfrac{[(m-1)(b-a)]}{(b+ma)(a+mb)}$

=

$\dfrac{(m^2-1)ab(b-a)}{m(a^2+b^2)+(m^2+1)ab} = \dfrac{(m^2-1)ab(b-a)}{m^2-1}$ [From (i)]

= $ab(b-a)$.

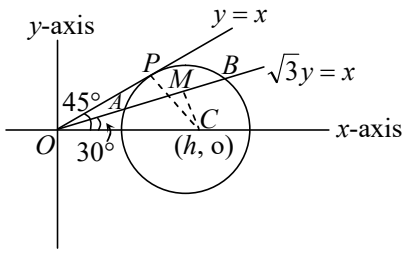
78. B

Sol. Mid-point of (1, 2) and (3, 6) is (2, 4), line passing through (1, 6) and (2, 4) is parallel to axis of the parabola (property). So, slope of its axis is –2 or slope of its directrix is 1/2, also in this case tangents are perpendicular. So, (1, 6) must lie on directrix,

$\therefore\ y - 6 = \dfrac{1}{2}(x-1) \quad \Rightarrow \quad x - 2y + 11 = 0$

79. A

Sol. $OP = CP = \dfrac{h}{\sqrt{2}} = AC$



$CM = h\sin 30° = \dfrac{h}{2}$

From Δ *ACM*

$AC^2 = AM^2 + CM^2$

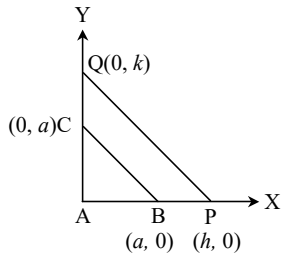
$\dfrac{h^2}{2} = 1 + \dfrac{h^2}{4} \Rightarrow \dfrac{h^2}{4} = 1 \quad \Rightarrow h = 2$

radius $= \sqrt{2}$

$(x-2)^2 + y^2 = 2\,; x^2 + y^2 - 4x + 2 = 0$

80. C

Sol. We take *A* as the origin and *AB* and *AC* as *x*-axis and *y*-axis respectively. Let *AP* = *h*, *AQ* = *k* Equation of the line *PQ* is



$$\frac{x}{h}+\frac{y}{k}=1 \qquad \ldots(i)$$

Given, $BP \cdot CQ = AB^2$

$\Rightarrow (h-a)(k-a)=a^2$

$\Rightarrow hk-ak-ah+a^2=a^2$ or $ak+ha=hk$

Or $\frac{a}{h}+\frac{a}{k}=1$ ....(ii)

From (ii), it follows that line (i) i.e. PQ always passes through a fixed point (*a*, *a*).

SECTION – B

81. 00008.00

Sol. Above expression is $\sum\left(2x^2+(1-x)^2\right)$, $x \in [0, 1]$

$x \in (p, q, r, s)$

Consider $f(x) = 2x^2 + (1 - x)^2$

82. 00000.00

Sol. Given

$f(x) \geq 0 \forall x \geq 0$

$f'(x)\cos^x - f(x)\sin x \leq 0$

$\frac{d}{dx}(f(x)\cos x) \leq 0$

$\therefore \int_{\frac{\pi}{2}}^{\frac{5\pi}{3}} \frac{d}{dx}(f(x)\cos x) \leq 0$

$\therefore\ f\left(\frac{5\pi}{3}\right)\cos\frac{5\pi}{3} \leq 0$

$\Rightarrow f\left(\frac{5\pi}{3}\right)=0$

83. 00002.00

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\frac{{}^nC_r}{n^r(r+3)} = \lim\limits_{n\to\infty}\left(\sum\limits_{r=0}^{n}\frac{{}^nC_r}{n^r}\int_0^1 x^{r+2}dx\right)$

$= \int_0^1\left(x^2\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}{}^nC_r\left(\frac{x}{n}\right)^r\right)dx = \int_0^1\left(x^2\lim\limits_{n\to\infty}\left(1+\frac{x}{n}\right)^n\right)dx$

$= \int_0^1 x^2e^x dx = e-2.$

84. 00001.00

Sol. As $f([i])=\frac{1}{1+[i]} \Rightarrow f(0)=1, f(1)=\frac{1}{2}, f(2)=\frac{1}{3}, f(3)=\frac{1}{4}, f(4)=\frac{1}{5}$

$\Rightarrow$ g(x) = (1+x) f(x)–1, is a polynomial of degree 5 with roots 0, 1, 2, 3, 4

$\Rightarrow g(x) = Ax(x-1)(x-2)(x-3)(x-4) \Rightarrow A=\frac{1}{\lfloor 5}$

$\Rightarrow (1+x)f(x)-1=\frac{1}{5!}x(x-1)(x-2)(x-3)(x-4)$.

85. 00005.00

Sol. $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}$

$\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 5\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 9\\ 0 & 1\end{bmatrix}$

$\begin{bmatrix}1 & 9\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 7\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 16\\ 0 & 1\end{bmatrix}$

If n is no of matrices that are multiplied then product is $\begin{bmatrix}1 & n^2\\ 0 & 1\end{bmatrix}$

$\therefore$ r = 2500.

86. 00003.00

Sol. $a^2+1\geq 2a,\ b^2+1\geq 2b,\ c^2+1\geq 2c$

$\Rightarrow \frac{a^2+1}{b+c}+\frac{b^2+1}{c+a}+\frac{c^2+1}{a+b}\geq \frac{2a}{a+b}+\frac{2b}{c+a}+\frac{2c}{a+b}$

Let b + c = x, c + a = y, a + b = z

$\Sigma\frac{2a}{b+c}=\Sigma\frac{y+z-x}{x}=\Sigma\left(\frac{x}{y}+\frac{y}{x}\right)-3\geq 6-3=3.$

87. 00002.00

Sol. We have, $a_n = S_n - S_{n-1}$ ...(i)

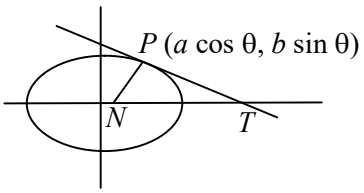
and $a_{n-1} = S_{n-1} - S_{n-2}$ ...(ii)

$\therefore d = a_n - a_{n-1} = (S_n - S_{n-1}) - (S_{n-1} - S_{n-2})$ [From (i) and (ii)]

$= S_n - 2S_{n-1} + S_{n-2}$

88. 00018.00

Sol. $e=\sqrt{1-\frac{b^2}{a^2}}=\frac{3}{5}$



$T\equiv\left(\frac{a}{\cos\theta},0\right),\ N\equiv\left(\frac{a^2-b^2}{a\sec\theta},0\right)$

On solving we get, $\cos\theta=\frac{1}{2},\ \sin\theta=\frac{\sqrt{3}}{2}$

$\therefore$ Area of $\Delta PSS' = 6\sqrt{3}$ sq. units

89. 00007.00

Sol. Deviation does not changes by changing a data

90. 00025.00

Sol. Let $I = f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right) = \int_0^{\pi/4} \log_e \left(\frac{1+\frac{1}{2}\tan z}{1+\frac{1}{3}\tan z}\right) dz = \int_0^{\pi/4} \log_e \left(\frac{3}{2}.\frac{2+\tan z}{3+\tan z}\right) dz$

$$= \int_0^{\pi/8} \left(\log_e \left(\frac{3}{2}.\frac{2+\tan z}{3+\tan z}\right) + \log_e \left(\frac{3}{2}.\frac{2+\tan\left(\frac{\pi}{4}-z\right)}{3+\tan\left(\frac{\pi}{4}-z\right)}\right)\right) dz = \frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$$

$$\Rightarrow \ I = \frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$$